Supplementary material

Table S1. Guarantee by data assumption

Each cell gives a representative method, the assumption it needs, and the price it pays. “Open” means no method in the reviewed set covers that cell.

guarantee	exchangeable	covariate shift	non-exchangeable / temporal	known groups
marginal coverage	Vovk et al. (2005), Lei et al. (2018), Angelopoulos & Bates (2023). One calibration pass.	Tibshirani et al. (2019). Likelihood ratio known or estimable.	Barber et al. (2023), Oliveira et al. (2022). Gap bounded by the violation.	Vovk (2012). Per-group quantiles; small n per group.
group-conditional coverage	Vovk (2012), Gibbs et al. (2025). Group-wise quantiles; smaller effective n.	open	open	Vovk (2012), Gibbs et al. (2025). Group labels known at test.
pointwise conditional coverage	Barber et al. (2021), Hore & Barber (2025). Larger sets.	open	open	open
risk control	Bates et al. (2021), Angelopoulos et al. (2022), Angelopoulos et al. (2025). Monotone loss; grid and multiplicity.	open in the reviewed set	Farinhas et al. (2023). Bounded non-exchangeability.	Combes et al. (2026).

Table S2. A calibrated threshold is split conformal

digits, alpha = 0.10, n_cal = 449, 5 seeds. Coverage is the mean over seeds. Raw keeps empty sets; forced fills them with the argmax. Size is mean cardinality. Pointwise set equality between the conformal threshold and split conformal: mismatches = 0.

strength = 0.05 (n_train = 44)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.294	0.833	0.299	1.000	0.701
threshold at max-prob quantile	0.800	0.837	0.920	1.006	0.086
threshold at true-class quantile	0.918	0.918	1.389	1.390	0.000
threshold at t = 1-q (conformal)	0.919	0.919	1.396	1.397	0.000
split conformal, naive score	0.919	0.919	1.396	1.397	0.000
split conformal, APS	0.917	0.952	1.886	1.922	0.036

strength = 0.25 (n_train = 224)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.713	0.938	0.719	1.000	0.281
threshold at max-prob quantile	0.886	0.938	0.912	1.000	0.088
threshold at true-class quantile	0.909	0.939	0.949	1.002	0.053
threshold at t = 1-q (conformal)	0.911	0.939	0.951	1.002	0.051

method	coverage raw	coverage forced	size raw	size forced	empty raw
split conformal, naive score	0.911	0.939	0.951	1.002	0.051
split conformal, APS	0.904	0.976	1.178	1.251	0.073

strength = 1.00 (n_train = 898)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.855	0.964	0.860	1.000	0.140
threshold at max-prob quantile	0.901	0.964	0.912	1.000	0.088
threshold at true-class quantile	0.908	0.964	0.920	1.000	0.080
threshold at t = 1-q (conformal)	0.910	0.964	0.922	1.000	0.078
split conformal, naive score	0.910	0.964	0.922	1.000	0.078
split conformal, APS	0.903	0.982	1.034	1.113	0.079

Table S3. Non-empty post-processing inflates coverage

digits, alpha = 0.10, 5 seeds. Delta = forced coverage - raw coverage. Raw empty is the share of sets that were empty before the argmax was forced in.

strength = 0.05 (n_train = 44)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.932	0.932	+0.000	0.000	1.565	1.565
naive	225	0.929	0.929	+0.000	0.000	1.491	1.491
naive	449	0.919	0.919	+0.000	0.000	1.396	1.397
APS randomized	100	0.914	0.954	+0.040	0.040	1.983	2.024
APS randomized	225	0.918	0.952	+0.033	0.034	1.864	1.898
APS randomized	449	0.920	0.954	+0.033	0.034	1.922	1.956
APS deterministic	100	0.893	0.981	+0.088	0.090	3.551	3.641
APS deterministic	225	0.899	0.984	+0.085	0.086	3.619	3.705
APS deterministic	449	0.898	0.981	+0.084	0.084	3.549	3.634
RAPS	100	0.922	0.952	+0.030	0.031	1.941	1.972
RAPS	225	0.917	0.951	+0.034	0.034	1.874	1.908
RAPS	449	0.911	0.945	+0.034	0.034	1.781	1.816

strength = 0.25 (n_train = 224)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.922	0.947	+0.025	0.044	0.981	1.025
naive	225	0.912	0.941	+0.029	0.050	0.956	1.007
naive	449	0.911	0.939	+0.028	0.051	0.951	1.002
APS randomized	100	0.911	0.979	+0.068	0.068	1.236	1.305

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
APS randomized	225	0.901	0.975	+0.074	0.074	1.193	1.268
APS randomized	449	0.919	0.978	+0.059	0.060	1.239	1.299
APS deterministic	100	0.892	0.995	+0.104	0.105	3.220	3.325
APS deterministic	225	0.907	0.996	+0.089	0.090	3.372	3.463
APS deterministic	449	0.898	0.995	+0.097	0.099	3.266	3.364
RAPS	100	0.910	0.975	+0.065	0.065	1.215	1.280
RAPS	225	0.894	0.973	+0.080	0.080	1.186	1.266
RAPS	449	0.896	0.973	+0.077	0.077	1.148	1.226

strength = 1.00 (n_train = 898)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.929	0.964	+0.035	0.056	0.944	1.000
naive	225	0.917	0.964	+0.047	0.071	0.929	1.000
naive	449	0.910	0.964	+0.054	0.078	0.922	1.000
APS randomized	100	0.903	0.984	+0.081	0.081	1.049	1.130
APS randomized	225	0.899	0.981	+0.082	0.082	1.044	1.126
APS randomized	449	0.913	0.983	+0.070	0.070	1.068	1.137
APS deterministic	100	0.894	1.000	+0.106	0.106	3.011	3.117
APS deterministic	225	0.908	1.000	+0.092	0.092	3.127	3.218
APS deterministic	449	0.911	1.000	+0.089	0.089	3.110	3.199
RAPS	100	0.900	0.982	+0.081	0.082	1.028	1.110
RAPS	225	0.887	0.982	+0.095	0.096	1.018	1.113
RAPS	449	0.884	0.980	+0.096	0.096	1.001	1.098

Table S4. The alpha_t update is not ACI

digits, alpha = 0.10, 5 seeds. Coverage and size are means over seeds. Min/max block are computed on the seed-averaged block curve. Clip is the share of steps at a clip bound.

schedule = linear

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.666	0.100	1.524	0.388	0.968	0.1000	0.000	0.000
fixed	0.005	0.832	0.023	3.371	0.776	0.948	0.0054	0.045	0.130
fixed	0.010	0.840	0.030	3.598	0.768	0.932	0.0064	0.071	0.139
fixed	0.020	0.836	0.038	3.444	0.732	0.928	0.0074	0.091	0.108
fixed	0.050	0.818	0.049	3.469	0.684	0.916	0.0090	0.132	0.105
fixed	0.100	0.804	0.046	3.182	0.668	0.908	0.0090	0.156	0.087
fixed	0.200	0.775	0.043	3.103	0.624	0.904	0.0170	0.203	0.098
buggy	0.005	0.849	0.025	3.773	0.784	0.940	0.0010	0.052	0.176
buggy	0.010	0.865	0.031	4.273	0.780	0.924	0.0010	0.061	0.216

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
buggy	0.020	0.869	0.036	4.580	0.752	0.944	0.0011	0.065	0.244
buggy	0.050	0.870	0.043	5.169	0.792	0.956	0.0012	0.070	0.297
buggy	0.100	0.781	0.205	4.868	0.668	0.868	0.0014	0.076	0.279
buggy	0.200	0.769	0.153	4.788	0.476	0.900	0.0019	0.091	0.260

schedule = step

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.663	0.085	1.488	0.432	0.960	0.1000	0.000	0.000
fixed	0.005	0.830	0.024	3.358	0.576	0.948	0.0064	0.051	0.130
fixed	0.010	0.835	0.028	3.492	0.680	0.940	0.0102	0.077	0.132
fixed	0.020	0.828	0.039	3.349	0.732	0.928	0.0098	0.102	0.104
fixed	0.050	0.808	0.046	3.176	0.712	0.916	0.0120	0.138	0.088
fixed	0.100	0.790	0.047	3.013	0.684	0.916	0.0190	0.166	0.079
fixed	0.200	0.767	0.042	3.010	0.664	0.900	0.0170	0.208	0.084
buggy	0.005	0.836	0.028	3.575	0.564	0.944	0.0010	0.060	0.161
buggy	0.010	0.845	0.031	3.826	0.676	0.924	0.0011	0.072	0.178
buggy	0.020	0.839	0.040	3.928	0.712	0.924	0.0012	0.076	0.188
buggy	0.050	0.809	0.067	4.078	0.636	0.892	0.0016	0.078	0.202
buggy	0.100	0.748	0.183	4.042	0.620	0.852	0.0015	0.086	0.199
buggy	0.200	0.733	0.163	4.262	0.484	0.840	0.0024	0.092	0.214

schedule = abrupt

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.739	0.042	1.453	0.360	0.960	0.1000	0.000	0.000
fixed	0.005	0.831	0.021	2.411	0.508	0.948	0.0054	0.040	0.070
fixed	0.010	0.840	0.023	2.574	0.608	0.940	0.0064	0.061	0.073
fixed	0.020	0.836	0.029	2.499	0.640	0.928	0.0074	0.080	0.062
fixed	0.050	0.824	0.033	2.425	0.644	0.916	0.0090	0.103	0.052
fixed	0.100	0.816	0.030	2.398	0.620	0.916	0.0090	0.124	0.048
fixed	0.200	0.796	0.022	2.528	0.612	0.900	0.0170	0.167	0.061
buggy	0.005	0.828	0.024	2.448	0.492	0.944	0.0010	0.042	0.079
buggy	0.010	0.832	0.039	2.598	0.588	0.924	0.0010	0.052	0.085
buggy	0.020	0.807	0.065	2.580	0.564	0.924	0.0011	0.054	0.089
buggy	0.050	0.767	0.107	2.904	0.636	0.892	0.0012	0.062	0.115
buggy	0.100	0.628	0.320	2.500	0.492	0.852	0.3387	0.048	0.092
buggy	0.200	0.576	0.259	2.649	0.316	0.776	0.0019	0.066	0.122

CIFAR-10, audit A: a calibrated threshold is split conformal

alpha = 0.10, n_cal = 2000, 5 seeds, 10k official test set. Coverage is the mean over seeds. Raw keeps empty sets; forced fills them with the argmax.

n_train	method	coverage raw	coverage forced	size raw	empty raw
100	raw threshold 0.90	0.290	0.692	0.310	0.690
100	max-prob quantile	0.668	0.702	0.921	0.103
100	true-class quantile	0.898	0.898	2.086	0.000
100	t = 1-q	0.898	0.898	2.090	0.000
100	split conformal	0.898	0.898	2.090	0.000
100	APS	0.896	0.911	2.471	0.016

n_train	method	coverage raw	coverage forced	size raw	empty raw
100	**accuracy**	**0.692**			
1000	raw threshold 0.90	0.647	0.803	0.705	0.295
1000	max-prob quantile	0.762	0.803	0.899	0.101
1000	true-class quantile	0.894	0.894	1.344	0.000
1000	t = 1-q	0.894	0.894	1.346	0.000
1000	split conformal	0.894	0.894	1.346	0.000
1000	APS	0.892	0.911	1.497	0.021
1000	**accuracy**	**0.803**			
5000	raw threshold 0.90	0.731	0.818	0.819	0.181
5000	max-prob quantile	0.771	0.818	0.892	0.108
5000	true-class quantile	0.899	0.899	1.286	0.000
5000	t = 1-q	0.899	0.899	1.287	0.000
5000	split conformal	0.899	0.899	1.287	0.000
5000	APS	0.897	0.907	1.351	0.011
5000	**accuracy**	**0.818**			
30000	raw threshold 0.90	0.716	0.855	0.756	0.244
30000	max-prob quantile	0.810	0.855	0.899	0.101
30000	true-class quantile	0.888	0.888	1.089	0.001
30000	t = 1-q	0.888	0.888	1.090	0.000
30000	split conformal	0.888	0.888	1.090	0.000
30000	APS	0.887	0.927	1.258	0.041
30000	**accuracy**	**0.855**			

CIFAR-10, audit B: non-empty post-processing inflates coverage

alpha = 0.10, n_cal = 2000, 5 seeds, 10k official test set. Delta = forced coverage - raw coverage. Raw empty is the share of empty sets.

n_train	score	coverage raw	coverage forced	delta	empty raw	size raw	size forced
100	naive	0.898	0.898	+0.000	0.000	2.090	2.090
100	APS randomized	0.898	0.913	+0.015	0.015	2.482	2.497
100	APS deterministic	0.904	0.972	+0.068	0.069	4.437	4.506
100	RAPS	0.899	0.911	+0.012	0.013	2.403	2.415
1000	naive	0.894	0.894	+0.000	0.000	1.346	1.346
1000	APS randomized	0.896	0.914	+0.018	0.019	1.515	1.534
1000	APS deterministic	0.902	0.993	+0.091	0.091	4.654	4.745
1000	RAPS	0.897	0.913	+0.016	0.016	1.490	1.506
5000	naive	0.899	0.899	+0.000	0.000	1.287	1.287
5000	APS randomized	0.900	0.909	+0.009	0.010	1.362	1.372
5000	APS deterministic	0.994	0.994	+0.000	0.000	6.083	6.083
5000	RAPS	0.899	0.907	+0.008	0.008	1.338	1.346
30000	naive	0.888	0.888	+0.000	0.000	1.090	1.091
30000	APS randomized	0.893	0.930	+0.036	0.037	1.271	1.309

n_train	score	coverage raw	coverage forced	delta	empty raw	size raw	size forced
30000	APS deterministic	0.899	0.994	+0.095	0.095	3.967	4.061
30000	RAPS	0.894	0.929	+0.035	0.036	1.261	1.297

Table S7. CIFAR-10-C style drift

Five corruption types (gaussian_noise, defocus_blur, contrast, brightness, jpeg_compression), three severities (1, 3, 5), 1,000 images per block, 3,000 points per stream, 3 seeds, alpha = 0.10, gamma = 0.02. Calibration is clean.

update	coverage	mean size	block sev 1	block sev 3	block sev 5	alpha final	clip	sets >= 9
static	0.451	1.30	0.588	0.457	0.308	—	—	0.000
fixed	0.828	4.93	0.853	0.829	0.802	0.0106	0.13	0.145
buggy	0.936	6.97	0.940	0.939	0.928	0.0019	0.05	0.380

Per corruption

Coverage and mean set size, averaged over three seeds.

corruption	static	fixed	buggy
gaussian_noise	0.22 / 1.34	0.66 / 5.45	0.81 / 7.02
defocus_blur	0.24 / 1.37	0.85 / 7.52	0.98 / 9.42
contrast	0.52 / 1.33	0.87 / 4.94	0.98 / 7.42
brightness	0.80 / 1.15	0.90 / 1.88	0.94 / 3.70
jpeg_compression	0.47 / 1.32	0.86 / 4.87	0.97 / 7.28

The incorrect update reports 10.8 points more coverage than the correct one, at a 41 percent larger mean set size, and raises the share of near-trivial sets from 14.5 to 38.0 percent. The correct update also falls short of 0.90, at 0.828.

Table S8. Library behavior audit

Versions checked on 2026-11-06. Line numbers refer to the named files in the package versions listed; the source snapshots were downloaded locally for the audit. Runtime: MAPIE, puncc, crepes, nonconformist, and conformal. Source inspection: TorchCP, coverforest, and AdaptiveConformal.

library	version	task	empty sets possible	default policy	evidence	ACI update
MAPIE	1.6.0	classification	LAC yes; APS policy-dependent	APS default include_last_label =True: no empty sets, coverage 0.9457 vs 0.90	classification.py:238-283; aps.py:355-358; lac.py:205-207; results/mapie_empty.json	fixed alpha_0 (time_series_regression.py:329)
TorchCP	1.2.1	classification	yes	no forcing; boolean mask (scores <= q_hat)	classification/predictor/base.py:111	fixed alpha (regression/predictor/aci.py:197)
puncc	0.9.3	classification	yes	no forcing; index lists, empty when no class passes	api/prediction_sets.py:70-78, 112-119	not implemented
crepes	0.9.1	classification	yes	no forcing; boolean mask (p_values >= 1-confidence)	base.py:340	not implemented

library	version	task	empty sets possible	default policy	evidence	ACI update
nonconformist	2.1.0	classification	yes	no forcing; boolean matrix of p-values > significance; smoothing=True	icp.py:199-250	not implemented
conformal	0.1	classification	yes (empty list possible)	no forcing; per-label thresholds at 100 - epsilon percentile	**init**.py:76-93, 95-108	not implemented
coverforest	0.0.3	classification	yes by default	allow_empty_sets =True default; False avoids empty sets; docs warn to use it sparingly	_forest.py:1457, 2006-2008	not applicable
AdaptiveConformal	GitHub main	regression intervals	not applicable	not applicable	R/aci_algorithm.R:78-81	fixed alpha (correct)

MAPIE 1.6.0, five seeds, alpha = 0.10

Synthetic five-class data, logistic regression, 2,000 test points per seed. The correct randomised APS is the last row. The default is the second row.

score	include_last_label	coverage (mean +/- SD)	empty rate	size
LAC	True	0.9016 +/- 0.0080	0.0000	3.151
APS, default	True	0.9457 +/- 0.0107	0.0000	3.676
APS	False	0.8284 +/- 0.0150	0.0000	2.684
APS, randomized	randomized	0.8998 +/- 0.0105	0.0004	3.232

The default APS setting over-covers by 4.6 points with zero empty sets. The randomised setting, which is the exact conformal rule, lands on the target. The difference is the non-empty post-processing, not the score.

Runtime cross-check

Same synthetic five-class task for all libraries: 8,000 samples, logistic regression, 2,000 test points, alpha = 0.10, test accuracy 0.487. One seed for every library in this table; the MAPIE numbers in the table above use five seeds. The puncc randomized row has no fixed seed; its empty rate was 0.0 to 0.2 percent across runs.

library	setting	coverage	empty rate	mean size
MAPIE	LAC	0.895	0.000	2.87
MAPIE	APS, include_last_label=True (default)	0.949	0.000	3.46
MAPIE	APS, include_last_label=False	0.809	0.000	2.48
MAPIE	APS, randomized	0.892	0.001	3.02
puncc	LAC	0.895	0.000	2.87
puncc	APS, rand=True	0.889	0.000	2.99
puncc	APS, rand=False	0.977	0.000	4.06
crepes	LAC (hinge), p >= alpha	0.895	0.000	2.87
nonconformist	ICP margin, smoothing=True	0.890	0.000	3.50
conformal	per-label thresholds, epsilon=10	0.890	0.000	3.02

coverforest 0.0.3 could not be run on scikit-learn 1.9: its forest code calls _get_n_samples_bootstrap without the sample_weight argument that the installed scikit-learn requires. TorchCP 1.2.1 was not run: its optional graph dependencies (torch_geometric) are not installed. Both were read at the source level.

Table S9. Coverage against four surveys

Authors' assessment of topical coverage, not of quality. A circle means the topic is not a focus of that survey, not that the survey is deficient. Symbols: full circle = comprehensive, half circle = partial, open circle = not a focus.

dimension	this paper	Angelopoulos & Bates 2021	Fontana et al. 2023	Campos et al. 2024	Zhou et al. 2025
foundations	half	full	full	half	full
scores	full	full	half	full	full
conditional coverage	full	half	full	half	half
risk control	full	half	half	half	half
non-exchangeable	full	open	half	half	half
language models	half	open	open	full	half
implementation audit	full	open	open	open	open
two-dataset experiments	full	open	open	open	open

The last two rows are the reason this paper exists. The first six rows are the reason it has to cite the other four.

Table S10. Experimental grid

dataset	train pool	strengths	calibration	test	model	scores	alpha	seeds
digits	898	44 / 224 / 898	449 for A and C; 100 / 225 / 449 for B	450	standardiser + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5
CIFAR-10	30,000	100 / 1,000 / 5,000 / 30,000	2,000 for A; 500 / 2,000 / 10,000 for B; 2,000 for C	10,000	ResNet-18 features + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5 for A and B; 3 for C
covertype	120,000-row subset	6,000 / 60,000	20,000	40,000	standardiser + logistic regression	naive, APS randomized, APS deterministic	0.05 / 0.10 / 0.20	5

The calibration column gives the sizes used in each audit. The strength column gives the number of training points used to fit the model; the calibration and test sets are disjoint from it.

Table S11. Covertype audit B

120,000-row stratified subset, 54 features, 7 classes, logistic regression, 20,000 calibration points, 40,000 test points, alpha = 0.10, 5 seeds.

train	score	raw coverage	raw empty	raw size	forced coverage	forced size	delta
6000	naive	0.900	0.000	1.53	0.900	1.53	0.000
6000	APS randomized	0.901	0.003	1.67	0.904	1.68	0.003

train	score	raw coverage	raw empty	raw size	forced coverage	forced size	delta
6000	APS deterministic	0.900	0.004	1.92	0.903	1.92	0.004
60000	naive	0.901	0.000	1.51	0.901	1.51	0.000
60000	APS randomized	0.900	0.003	1.65	0.903	1.66	0.003
60000	APS deterministic	0.899	0.001	1.89	0.900	1.89	0.001

The two training sizes give test accuracies of 0.717 and 0.721.

delta is forced coverage minus raw coverage. The identity delta = empty rate x conditional accuracy holds. The empty rate is 0.0 to 0.4 percent, so the change is 0.0 to 0.4 points. The trap is a regime effect, not a universal constant.

Table S12. Cost-sensitive reading of the fallback

At alpha = 0.10, over cells with a non-zero raw empty rate. a_E is the conditional accuracy of the inserted label on empty sets, estimated as delta / empty rate. The threshold is the cost ratio c_a / c_e above which forcing is better than abstaining.

dataset	score	cells	mean empty rate	mean a_E	threshold c_a/c_e
digits	aps_rand	45	0.060	0.99	0.01
digits	aps_det	45	0.093	0.99	0.01
cifar10	aps_rand	60	0.020	0.96	0.04
cifar10	aps_det	45	0.084	0.99	0.01
covertype	aps_rand	10	0.003	0.94	0.06
covertype	aps_det	10	0.003	0.94	0.06

The threshold is small in every dataset (0.01 to 0.06); the empty rate is not. The expected loss difference at a given cost ratio is proportional to the empty rate, so the same rule matters much more on digits than on covertype.

Reporting checklist

1. Report the nominal level and the realized coverage, raw and forced.
2. Report the empty rate.
3. Report the calibration size and the finite-sample correction.
4. Report the score and the set-size distribution.
5. Report the update rule and its target.
6. Report the library version and its non-empty policy.